

计算系统生物学作业5

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1 Stability of a reaction diffusion system

Consider the following general reaction diffusion system

$$\frac{\partial u}{\partial t} = A_1 F(u, v) + D_1 \frac{\partial^2 u}{\partial x^2} \quad (1)$$

$$\frac{\partial v}{\partial t} = A_2 G(u, v) + D_2 \frac{\partial^2 v}{\partial x^2} \quad (2)$$

a.

证明. let $\bar{u} = u, \bar{v} = v$. We introduce L and T. Define $\bar{x} = \frac{x}{L}, \bar{t} = \frac{t}{T}$, then by equation 1 we have

$$\frac{1}{T} \frac{\partial \bar{u}}{\partial \bar{t}} = A_1 F(\bar{u}, \bar{v}) + \frac{D_1}{L^2} \frac{\partial^2 \bar{u}}{\partial \bar{x}^2} \quad (3)$$

Set $T = \frac{L^2}{D_1}, \gamma = TA_1 = \frac{A_1 L^2}{D_1}$, and define $f(\bar{u}, \bar{v}) = F(u, v)$, we have

$$\frac{\partial \bar{u}}{\partial \bar{t}} = \gamma f(\bar{u}, \bar{v}) + \frac{\partial^2 \bar{u}}{\partial \bar{x}^2} \quad (4)$$

For equation 2

$$\frac{\partial \bar{v}}{\partial \bar{t}} = TA_2 G + \frac{TD_2}{L^2} \frac{\partial^2 \bar{v}}{\partial \bar{x}^2} = \frac{A_2 L^2}{D_1} G + \frac{D_2}{D_1} \frac{\partial^2 \bar{v}}{\partial \bar{x}^2} \quad (5)$$

Define $g(\bar{u}, \bar{v}) = \frac{A_2}{A_1} G(u, v), d = \frac{D_2}{D_1}$, we have

$$\frac{\partial \bar{v}}{\partial \bar{t}} = \gamma g(\bar{u}, \bar{v}) + d \frac{\partial^2 \bar{v}}{\partial \bar{x}^2} \quad (6)$$

□

b. The homogeneous solution (u_0, v_0) satisfies that $f(u_0, v_0) = 0, g(u_0, v_0) = 0$. The Jacobi matrix $J = \gamma \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}$. For stable solutions, we require the real part of eigenvalues of J to be negative. This gives us

$$\text{tr}(J) = \gamma(f_u + g_v) < 0 \implies f_u + g_v < 0 \quad (7)$$

$$\det(J) = \gamma^2(f_u g_v - g_u f_v) > 0 \implies f_u g_v - g_u f_v > 0 \quad (8)$$

c. Consider inhomogeneous perturbations of the form

$$u(x, t) = u_0 + \delta u(t) \cos(x/l) \quad (9)$$

$$v(x, t) = v_0 + \delta v(t) \cos(x/l) \quad (10)$$

Notice that $\frac{\partial^2 u}{\partial x^2} = -\frac{1}{l^2} \delta u \cos(x/l)$.

By Taylor's expansion we have

$$f(u, v) \approx f(u_0, v_0) + f_u \delta u \cos(x/l) + f_v \delta v \cos(x/l) = f_u \delta u \cos(x/l) + f_v \delta v \cos(x/l) \quad (11)$$

Hence we have

$$\frac{d(\delta u)}{dt} = (\gamma f_u - \frac{1}{l^2}) \delta u + \gamma f_v \delta v \quad (12)$$

$$\frac{d(\delta v)}{dt} = \gamma g_u \delta u + (\gamma g_v - \frac{d}{l^2}) \delta v \quad (13)$$

d.

证明. Denote $k = 1/l^2$, we have $J_k = \begin{pmatrix} \gamma f_u - k^2 & \gamma f_v \\ \gamma g_u & \gamma g_v - dk^2 \end{pmatrix}$

We already know that

$$\text{tr}(J_k) = \gamma(f_u + g_v) - (1 + d)k^2 < 0 \quad (14)$$

Hence the following equation must hold:

$$\det(J_k) = dk^4 - \gamma(df_u + g_v)k^2 + \gamma^2(f_u g_v - f_v g_u) < 0 \quad (15)$$

Consider $h(x) = dx^2 - \gamma(df_u + g_v)x + \gamma^2(f_u g_v - f_v g_u)$, then $\det(J_k) = h(k^2)$.

Hence

$$\frac{\gamma(df_u + g_v)}{2d} > 0 \implies df_u + g_v > 0 \quad (16)$$

$$(-\gamma(df_u + g_v))^2 - 4d\gamma^2(f_u g_v - f_v g_u) > 0 \implies (df_u + g_v)^2 - 4d(f_u g_v - f_v g_u) > 0 \quad (17)$$

If $f_u < 0$ and $g_v < 0$, then $df_u + g_v < 0$. If $f_u > 0$ and $g_v > 0$, then $f_u + g_v > 0$, contradiction! Hence the signs of f_u and g_v must be different.

From equation 7, we know that

$$f_u + g_v < 0 \implies -g_v > f_u \quad (18)$$

From equation 16, we know that

$$df_u + g_v > 0 \implies df_u > -g_v \quad (19)$$

Hence $df_u > -g_v > f_u$. Since $f_u > 0$, we have $d > 1$. $\square$

e. Since f_u, g_v have different signs, and from equation 8, we know that f_v, g_u have different signs.

f_u	f_v	g_u	g_v	u	v	d
+	-	+	-	$u+, v+$	$u-, v-$	> 1
+	+	-	-	$u+, v-$	$u+, v-$	> 1
-	-	+	+	$u-, v+$	$u-, v+$	< 1
-	+	-	+	$u-, v-$	$u+, v+$	< 1

Compare with LALRI. Local activation is showed by $f_u > 0$, so locally a small perturbation will be amplified and form a local maximum. Long -range inhibition is showed by $d > 1$, which means the inhibitor diffuses faster than the activator.

$d > 1$ implies that $\frac{D_v}{D_u} > 1$, which means the inhibitor diffuses faster than activator, and this contributes to the formation of the pattern.

f. [1]Rüdiger, S., Nicola, E.M., Casademunt, J. and Kramer, L., 2007. Theory of pattern forming systems under traveling-wave forcing. Physics Reports, 447(3-6), pp.73-111.

$$\frac{\partial u}{\partial t} = a - cu - 4\frac{uv}{1+u^2} - \phi_0 - \xi \cos(k_f x - \omega t) + \frac{\partial^2 u}{\partial x^2} \quad (20)$$

$$\frac{\partial v}{\partial t} = \sigma(cu - \frac{uv}{1+u^2} + \phi_0 \xi \cos(k_f x - \omega t) + d\frac{\partial^2 v}{\partial x^2}) \quad (21)$$

where $a = 16, c = 0.6, d = 1.07, \sigma = 301, \xi = 0, \phi_0 = 2.3$, balance point $(u_0, v_0) = (1.5, 6.93)$. By calculating, we have $f_u = 2.68 > 0, f_v = -1.85 < 0, g_u = 427.46 > 0, g_v = -138.92 < 0$. This is local activation long range inhibition.

[2]Uecker, H. and Wetzel, D., 2014. Numerical Results for Snaking of Patterns over Patterns in Some 2D Selkov–Schnakenberg Reaction-Diffusion Systems. SIAM Journal on Applied Dynamical Systems, 13(1), pp.94-128.

$$\frac{\partial u}{\partial t} = -u + u^2v + b + av + \frac{\partial^2 u}{\partial x^2} \quad (22)$$

$$\frac{\partial v}{\partial t} = \lambda - u^2v - av + d\frac{\partial^2 v}{\partial x^2} \quad (23)$$

where $a = b = 0, d = 60, \lambda = 3.2805$, balance point $(u_0, v_0) = (3.2085, 0.3117)$. By calculating we have $f_u = 1.00 > 0, f_v = 10.29 > 0, g_u = -2.00 < 0, g_v = -10.29 < 0$, both steady.

2 Turing patterns in 2D

Consider the following dimensionless system restricted to a square region in 2D with a side of length L and periodic boundary conditions.

$$\frac{\partial u}{\partial t} = \gamma(a - u - \frac{\rho uv}{1 + u + Ku^2}) + \nabla^2 u \quad (24)$$

$$\frac{\partial v}{\partial t} = \gamma(\alpha(b - v) - \frac{\rho uv}{1 + u + Ku^2}) + d\nabla^2 v \quad (25)$$

a.

证明. Define $x' = x/L, y' = y/L, \nabla^2 = \frac{\partial^2}{\partial x'^2} + \frac{\partial^2}{\partial y'^2} = \frac{1}{L^2} \nabla'^2$. Denote $f(u, v) = a - u - \frac{\rho uv}{1 + u + Ku^2}, g(u, v) = \alpha(b - v) - \frac{\rho uv}{1 + u + Ku^2}$. Define $t' = t/L$.

The original equations are changed to

$$\frac{\partial u}{\partial t'} = L^2 \gamma f(u, v) + \nabla'^2 u \quad (26)$$

$$\frac{\partial v}{\partial t'} = L^2 \gamma g(u, v) + d\nabla'^2 v \quad (27)$$

Hence $(L, \gamma) \iff (1, \gamma')$, where $\gamma' = L^2 \gamma$. $\square$

b. To find homogeneous stationary solutions, we know that $f(u, v) = g(u, v) = 0$, hence by subtraction we have $v = b - \frac{a-u}{\alpha}$. For $a = 92, b = 64, \alpha = 1.5$, we have $v = \frac{4+u}{1.5}$. $\rho = 18.5, K = 0.1$, we notice that when $u_0 \approx 10, v_0 \approx 9.33$. The precise solution: $u_0 \approx 9.94, v_0 \approx 9.29$.

c.

证明. Now we calculate the Jacobi matrix. $f_u \approx -2.52, f_v \approx -8.83, g_u \approx -3.52, g_v \approx -10.33$. Hence $tr(J) \approx -7.81 < 0, det(J) \approx 5.05 > 0$. To show that the system is unstable to some inhomogeneous perturbations, we need to show (i) $df_u + g_v > 0$. Notice $d = 10, f_u = 2.52, g_v = -10.33$ satisfies (i); (ii) $(df_u + g_v)^2 > 4d \det(J)$. Notice $d = 10, f_u = 2.52, g_v = -10.33, \det(J) = 5.05$ satisfies (ii). $\square$

d. Unstable wave vector k satisfies positive real part

$$k^2 \in [\frac{\gamma}{2d}((df_u + g_v) - \sqrt{\Delta}), \frac{\gamma}{2d}((df_u + g_v) + \sqrt{\Delta})] \quad (28)$$

i.e. unstable wave vector $k \in [\sqrt{0.51\Delta}, \sqrt{0.98\Delta}]$

unstable wave length: $\lambda = \frac{2\pi}{k}$, approximately $[\frac{6.34}{\sqrt{\gamma}}, \frac{8.80}{\sqrt{\gamma}}]$.

e. For the proposed solution

$$u_{n,m_x,m_y} = \xi^n e^{i(m_x k_x + m_y k_y + y)\Delta l} \quad (29)$$

We have

$$\xi = 1 - \frac{4D\Delta t}{\Delta l^2} (\sin^2 \frac{k_x \Delta l}{2} + \sin^2 \frac{k_y \Delta l}{2}) \quad (30)$$

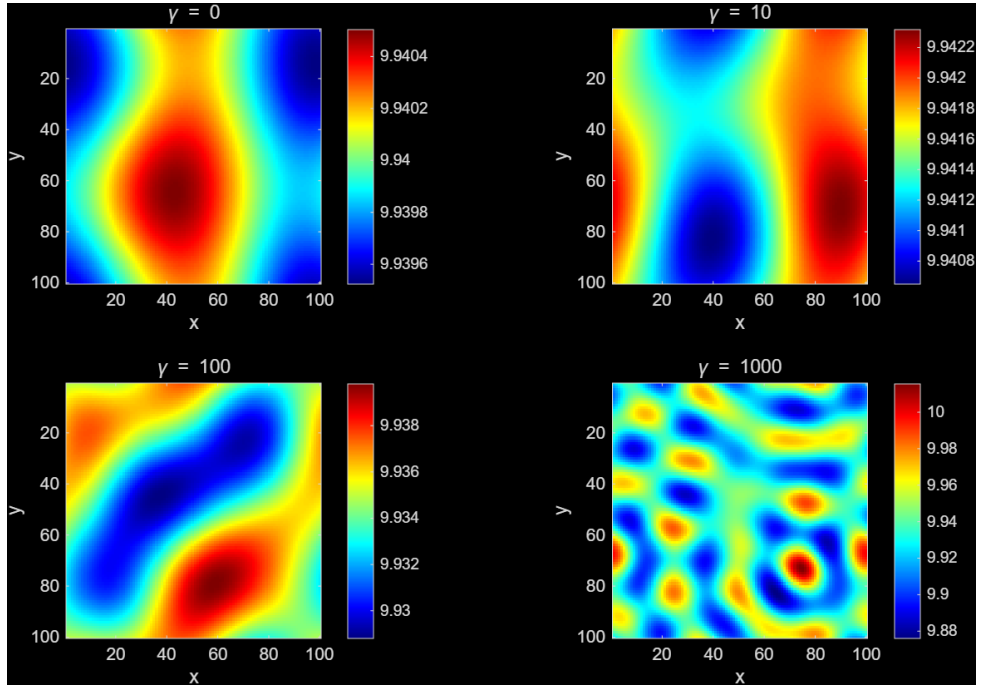
Here we require $|\xi| \leq 1, \forall k_x, k_y$. Hence $1 - \frac{4D\Delta t}{\Delta l^2} (1+1) \geq -1 \implies \delta t \leq \frac{\Delta l^2}{4D}$. To avoid vibration, we have $\Delta t < \frac{\Delta l^2}{4D}$.

If there are two species with different diffusion coefficients, we have

$$\Delta t < \frac{\Delta l^2}{4 \max(D_u, D_v)} \quad (31)$$

f. Under this specific parameter setting, the system tends to form spots. For different γ :

1. $\gamma = 0$: No patterns. The whole plane is completely even color.
2. $\gamma = 10$: No patterns usually. $\lambda \approx 2 > L = 1$, and the perturbation will be inhibited.
3. $\gamma = 100$: Clear patterns are observed.
4. $\gamma = 1000$: Detailed and complex patterns are observed.



To form a pattern, two basic conditions have to be satisfied:

- (i) Parameters at unstable areas
- (ii) $L > \lambda_c$

3 Spatial oscillations in E.coli - Howard et al model

Consider the Howard model for the spatial oscillations of the proteins that control the positioning of the midcell division plane

$$\frac{\partial \rho_D}{\partial t} = -\frac{\sigma_1 \rho_D}{1 + \sigma'_1 \rho_e} + \sigma_2 \rho_e \rho_d + D_D \frac{\partial^2 \rho_D}{\partial x^2} \quad (32)$$

$$\frac{\partial \rho_d}{\partial t} = +\frac{\sigma_1 \rho_D}{1 + \sigma'_1 \rho_e} - \sigma_2 \rho_e \rho_d \quad (33)$$

$$\frac{\partial \rho_E}{\partial t} = +\frac{\sigma_4 \rho_e}{1 + \sigma'_4 \rho_D} - \sigma_3 \rho_D \rho_E + D_E \frac{\partial^2 \rho_E}{\partial x^2} \quad (34)$$

$$\frac{\partial \rho_e}{\partial t} = -\frac{\sigma_4 \rho_e}{1 + \sigma'_4 \rho_D} + \sigma_3 \rho_D \rho_E \quad (35)$$

a.

1. diffusion: $D_D \frac{\partial^2 \rho_D}{\partial x^2}, D_E \frac{\partial^2 \rho_E}{\partial x^2}$. These show the free diffusion of MinD and MinE in the cytoplasm.
2. MinD on the plasma membrane: $+\frac{\sigma_1 \rho_D}{1 + \sigma'_1 \rho_e}$. This shows the speed at which MinD combines to the plasma membrane.
3. MinD off the plasma membrane: $-\sigma_2 \rho_e \rho_d$.
4. MinE on the plasma membrane: $+\sigma_3 \rho_D \rho_E$.
5. MinE off the plasma membrane: $-\frac{\sigma_4 \rho_e}{1 + \sigma'_4 \rho_D}$

b. For the total amount of MinD:

$$N_D = \int_0^L (\rho_D + \rho_d) dx \quad (36)$$

$$\frac{dN_D}{dt} = \int_0^L \left(\frac{\partial \rho_D}{\partial t} + \frac{\partial \rho_d}{\partial t} \right) dx = \int_0^L D_D \frac{\partial^2 \rho_D}{\partial x^2} dx = D_D \left[\frac{\partial \rho_D}{\partial x} \right]_0^L = 0 \quad (37)$$

Hence MinD is conserved. Similarly, MinE is conserved.

$\langle \rho_D \rangle$ and $\langle \rho_E \rangle$ enter the model as the initial value constraint/global parameters.

Based on Howard, M., Rutenberg, A.D. and de Vet, S., 2001. Dynamic compartmentalization of bacteria: accurate division in E. coli. Physical review letters, 87(27), p.278102., we estimate that $\langle \rho_D \rangle \approx 1500 \mu m^{-1}$, $\langle \rho_E \rangle \approx 85 \mu m^{-1}$

c. For trivial solution $\rho_D = 0, \rho_d = \langle \rho_D \rangle, \rho_E = \langle \rho_E \rangle, \rho_e = 0$, we consider linear stability under homogeneous perturbation.

$$\frac{d\delta \rho_D}{dt} \approx -\sigma_1 \delta \rho_D + \sigma_2 \langle \rho_D \rangle \delta \rho_e \quad (38)$$

$$\frac{d\delta \rho_e}{dt} \approx \sigma_3 \langle \rho_E \rangle \delta \rho_D - \sigma_4 \delta \rho_e \quad (39)$$

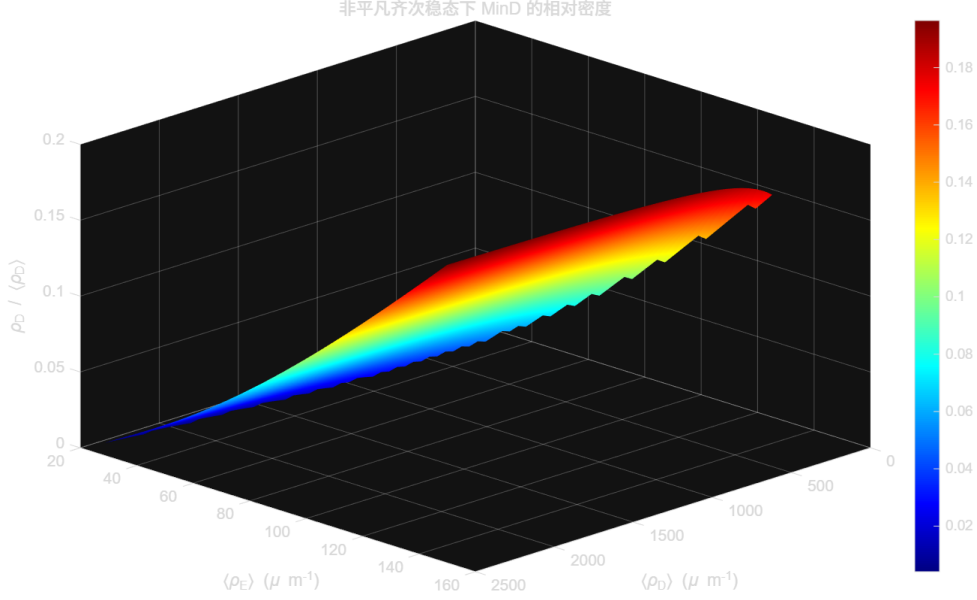
$$J = \begin{pmatrix} -\sigma_1 & \sigma_2 \langle \rho_D \rangle \\ \sigma_3 \langle \rho_E \rangle & -\sigma_4 \end{pmatrix} \quad (40)$$

Since $tr(J) = -\sigma_1 - \sigma_4 < 0$, we require that $\det(J) < 0$, i.e. $\sigma_1 \sigma_4 < \sigma_2 \sigma_3 \langle \rho_D \rangle \langle \rho_E \rangle$.

d. Since $\rho_d = \langle \rho_D \rangle - \rho_D, \rho_E = \langle \rho_E \rangle - \rho_e$, the steady state equations:

$$\frac{\sigma_1 \rho_D}{1 + \sigma_1' \rho_e} = \sigma_2 \rho_e (\langle \rho_d \rangle - \rho_D) \quad (41)$$

$$\frac{\sigma_4 \rho_e}{1 + \sigma_4' \rho_D} = \sigma_3 \rho_D (\langle \rho_E \rangle - \rho_e) \quad (42)$$



e. $\left. \frac{\partial \rho}{\partial x} \right|_{x=\pm L/2}$ For cosine perturbation, the derivative $-k\delta\rho \sin(kx)$, hence $\sin(k\frac{L}{2}) = 0, \sin(k\frac{-L}{2}) = 0$. This means $k = \frac{2n\pi}{L}$. These models are symmetric about the cell center.

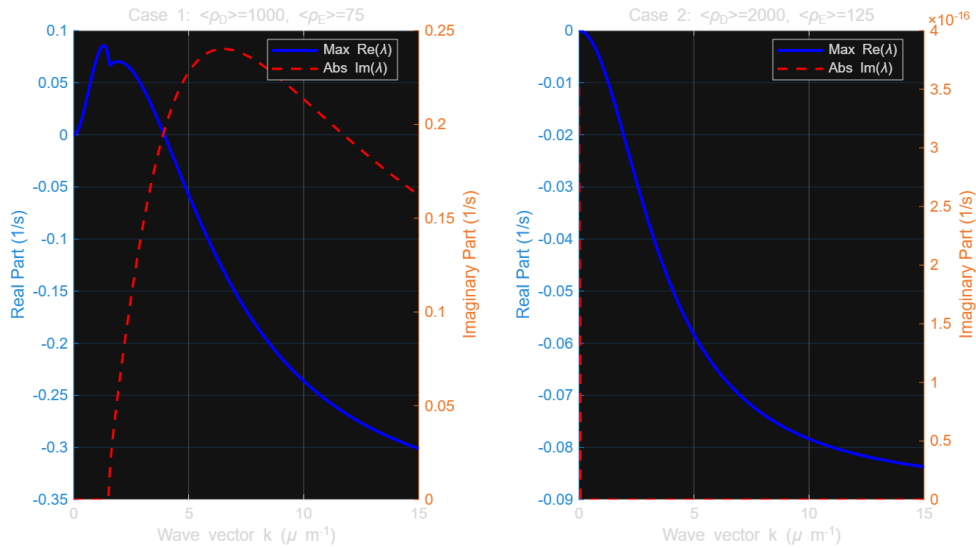
For sine perturbation, the derivative $k\delta\rho \cos(kx)$, $\cos(k\frac{L}{2}) = 0$, which means $k = \frac{(2n+1)\pi}{L}$. These models are antisymmetric about the cell center.

f.

$$D_{mat} = \text{diag}(D_0, 0, D_E, 0) \quad (43)$$

The evolution of the perturbation is determined by the eigenvalues of

$$M(k) = J - k^2 D_{mat} \quad (44)$$



g. stable: $\forall k > 0, \text{Re}\lambda_{max} < 0$

unstable: $\exists k, s.t. \text{Re}(\lambda) > 0$. If $\text{Im}(\lambda) = 0$: static pattern. Otherwise: traveling wave or standing wave oscillation

h.

$$\Delta t \leq \frac{\Delta x^2}{2D_{max}} \quad (45)$$

